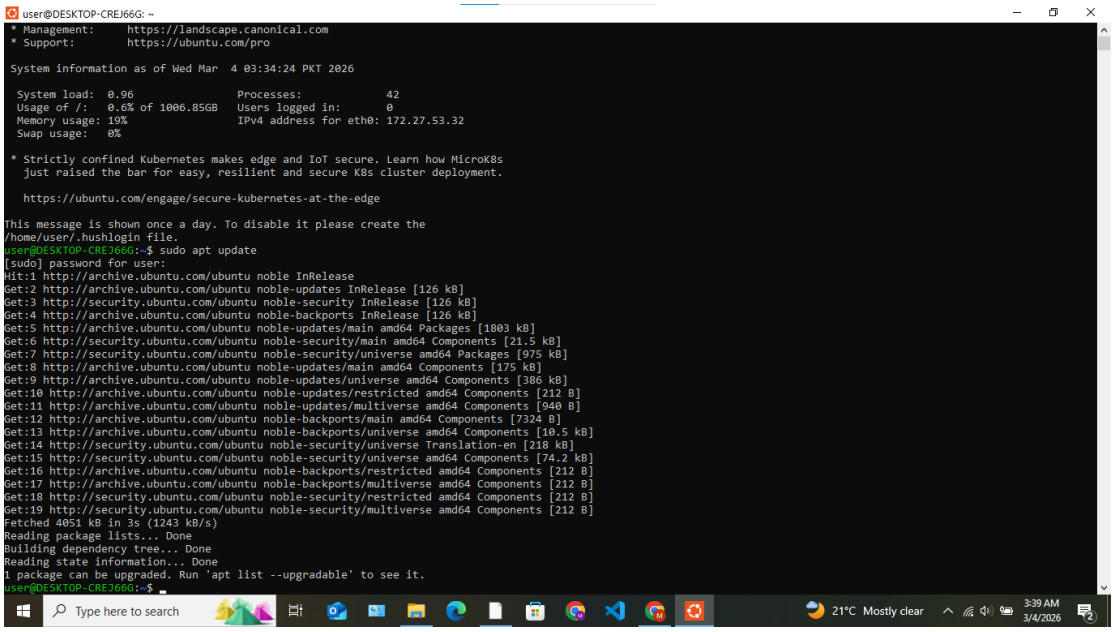
Muhammad Ali Husnain Butt

24k-874

OS-Theory: 01



user@DESKTOP-CREJ66G: ~/linux-6.19.5

```
  GNU nano 7.2                                                    Makefile
# SPDX-License-Identifier: GPL-2.0
VERSION = 6
PATCHLEVEL = 19
SUBLEVEL = 5
EXTRAVERSION =-k240874
NAME = Baby Opossum Posse

# *DOCUMENTATION*
# To see a list of typical targets execute "make help"
# More info can be located in ./README
# Comments in this file are targeted only to the developer, do not
# expect to learn how to build the kernel reading this file.

ifeq ($(filter output-sync,$(.FEATURES)),)
$(error GNU Make >= 4.0 is required. Your Make version is $(MAKE_VERSION))
endif

$(if $(filter __%, $(MAKECMDGOALS)), \
        $(error targets prefixed with '__' are only for internal use))

# That's our default target when none is given on the command line
PHONY := __all
__all:

# We are using a recursive build, so we need to do a little thinking
# to get the ordering right.
#
# Most importantly: sub-Makefiles should only ever modify files in
# their own directory. If in some directory we have a dependency on
# a file in another dir (which doesn't happen often, but it's often
# unavoidable when linking the built-in.a targets which finally
# turn into vmlinux), we will call a sub make in that other dir, and
# after that we are sure that everything which is in that other dir
# is now up to date.
#
# The only cases where we need to modify files which have global
# effects are thus separated out and done before the recursive
# descending is started. They are now explicitly listed as the
# prepare rule.

                                                        [ Read 2212 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket  M-Q Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was    M-W Next
```

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```
  GNU nano 7.2                                            kernel/Makefile
# SPDX-License-Identifier: GPL-2.0
#
# Makefile for the linux kernel.
#

obj-y     = fork.o exec_domain.o panic.o \
            cpu.o exit.o softirq.o resource.o \
            sysctl.o capability.o ptrace.o user.o \
            signal.o sys.o umh.o workqueue.o pid.o task_work.o \
            extable.o params.o \
            kthread.o sys_ni.o nsproxy.o nstree.o nscommon.o \
            notifier.o ksysfs.o cred.o reboot.o \
            async.o range.o smpboot.o ucount.o regset.o ksyms_common.o hello.o
obj-$(CONFIG_MULTIUSER) += groups.o
obj-$(CONFIG_VHOST_TASK) += vhost_task.o

ifdef CONFIG_FUNCTION_TRACER
# Do not trace internal ftrace files
CFLAGS_REMOVE_irq_work.o = $(CC_FLAGS_FTRACE)
endif

# Branch profiling isn't noinstr-safe
ifdef CONFIG_TRACE_BRANCH_PROFILING
CFLAGS_context_tracking.o += -DDISABLE_BRANCH_PROFILING
endif

# Prevents flicker of uninteresting __do_softirq()/__local_bh_disable_ip()
# in coverage traces.
KCOV_INSTRUMENT_softirq.o := n
# Avoid KCSAN instrumentation in softirq ("No shared variables, all the data
# are CPU local" => assume no data races), to reduce overhead in interrupts.
KCSAN_SANITIZE_softirq.o = n
# These are called from save_stack_trace() on slub debug path,
# and produce insane amounts of uninteresting coverage.
KCOV_INSTRUMENT_extable.o := n
KCOV_INSTRUMENT_stacktrace.o := n
# Don't self-instrument.
KCOV_INSTRUMENT_kcov.o := n
# If sanitizers detect any issues in kcov, it may lead to recursion
# via printk, etc.
                                                        [ Read 206 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket  M-Q Previous
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```

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GNU nano 7.2 include/linux/syscalls.h

```
static inline unsigned int ksys_personality(unsigned int personality)
{
        unsigned int old = current->personality;

        if (personality != 0xffffffff)
                set_personality(personality);

        return old;
}

/* for __ARCH_WANT_SYS_IPC */
long ksys_semtimedop(int semid, struct sembuf __user *tsops,
                     unsigned int nsops,
                     const struct __kernel_timespec __user *timeout);
long ksys_semget(key_t key, int nsems, int semflg);
long ksys_old_semctl(int semid, int semnum, int cmd, unsigned long arg);
long ksys_msgget(key_t key, int msgflg);
long ksys_old_msgctl(int msqid, int cmd, struct msqid_ds __user *buf);
long ksys_msgrcv(int msqid, struct msgbuf __user *msgp, size_t msgsz,
                 long msgtyp, int msgflg);
long ksys_msgsnd(int msqid, struct msgbuf __user *msgp, size_t msgsz,
                 int msgflg);
long ksys_shmget(key_t key, size_t size, int shmflg);
long ksys_shmdt(char __user *shmaddr);
long ksys_old_shmctl(int shmid, int cmd, struct shmid_ds __user *buf);
long compat_ksys_semtimedop(int semid, struct sembuf __user *tsems,
                            unsigned int nsops,
                            const struct old_timespec32 __user *timeout);
long __do_semtimedop(int semid, struct sembuf *tsems, unsigned int nsops,
                     const struct timespec64 *timeout,
                     struct ipc_namespace *ns);

int __sys_getsockopt(int fd, int level, int optname, char __user *optval,
                     int __user *optlen);
int __sys_setsockopt(int fd, int level, int optname, char __user *optval,
                     int optlen);
asmlinkage long sys_hello_world(void);
#endif
```

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GNU nano 7.2 arch/x86/entry/syscalls/syscall_64.tbl

```
455     common  futex_wait              sys_futex_wait
456     common  futex_requeue           sys_futex_requeue
457     common  statmount               sys_statmount
458     common  listmount               sys_listmount
459     common  lsm_get_self_attr       sys_lsm_get_self_attr
460     common  lsm_set_self_attr       sys_lsm_set_self_attr
461     common  lsm_list_modules        sys_lsm_list_modules
462     common  mseal                   sys_mseal
463     common  setxattrat              sys_setxattrat
464     common  getxattrat              sys_getxattrat
465     common  listxattrat             sys_listxattrat
466     common  removexattrat           sys_removexattrat
467     common  open_tree_attr          sys_open_tree_attr
468     common  file_getattr            sys_file_getattr
469     common  file_setattr            sys_file_setattr
470     common  listns                  sys_listns
471     common  hello_world             sys_hello_world
#
# Due to a historical design error, certain syscalls are numbered differently
# in x32 as compared to native x86_64.  These syscalls have numbers 512-547.
# Do not add new syscalls to this range.  Numbers 548 and above are available
# for non-x32 use.
#
512     x32     rt_sigaction            compat_sys_rt_sigaction
513     x32     rt_sigreturn            compat_sys_x32_rt_sigreturn
514     x32     ioctl                   compat_sys_ioctl
515     x32     readv                   sys_readv
516     x32     writev                  sys_writev
517     x32     recvfrom                compat_sys_recvfrom
518     x32     sendmsg                 compat_sys_sendmsg
519     x32     recvmsg                 compat_sys_recvmsg
520     x32     execve                  compat_sys_execve
521     x32     ptrace                  compat_sys_ptrace
522     x32     rt_sigpending           compat_sys_rt_sigpending
523     x32     rt_sigtimedwait         compat_sys_rt_sigtimedwait_time64
524     x32     rt_sigqueueinfo         compat_sys_rt_sigqueueinfo
525     x32     sigaltstack             compat_sys_sigaltstack
526     x32     timer_create            compat_sys_timer_create
527     x32     mq_notify               compat_sys_mq_notify
528     x32     kexec_load              compat_sys_kexec_load
```

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GNU nano 7.2 arch/x86/entry/syscalls/syscall_64.tbl

```
455     common  futex_wait              sys_futex_wait
456     common  futex_requeue           sys_futex_requeue
457     common  statmount               sys_statmount
458     common  listmount               sys_listmount
459     common  lsm_get_self_attr       sys_lsm_get_self_attr
460     common  lsm_set_self_attr       sys_lsm_set_self_attr
461     common  lsm_list_modules        sys_lsm_list_modules
462     common  mseal                   sys_mseal
463     common  setxattrat              sys_setxattrat
464     common  getxattrat              sys_getxattrat
465     common  listxattrat             sys_listxattrat
466     common  removexattrat           sys_removexattrat
467     common  open_tree_attr          sys_open_tree_attr
468     common  file_getattr            sys_file_getattr
469     common  file_setattr            sys_file_setattr
470     common  listns                  sys_listns
471     common  hello_world             sys_hello_world
#
# Due to a historical design error, certain syscalls are numbered differently
# in x32 as compared to native x86_64.  These syscalls have numbers 512-547.
# Do not add new syscalls to this range.  Numbers 548 and above are available
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#
512     x32     rt_sigaction            compat_sys_rt_sigaction
513     x32     rt_sigreturn            compat_sys_x32_rt_sigreturn
514     x32     ioctl                   compat_sys_ioctl
515     x32     readv                   sys_readv
516     x32     writev                  sys_writev
517     x32     recvfrom                compat_sys_recvfrom
518     x32     sendmsg                 compat_sys_sendmsg
519     x32     recvmsg                 compat_sys_recvmsg
520     x32     execve                  compat_sys_execve
521     x32     ptrace                  compat_sys_ptrace
522     x32     rt_sigpending           compat_sys_rt_sigpending
523     x32     rt_sigtimedwait         compat_sys_rt_sigtimedwait_time64
524     x32     rt_sigqueueinfo         compat_sys_rt_sigqueueinfo
525     x32     sigaltstack             compat_sys_sigaltstack
526     x32     timer_create            compat_sys_timer_create
527     x32     mq_notify               compat_sys_mq_notify
528     x32     kexec_load              compat_sys_kexec_load
```

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GNU nano 7.2 kernel/sys.c

```
                        s.mem_unit <<= 1;
                        bitcount++;
                }

                s.totalram >>= bitcount;
                s.freeram >>= bitcount;
                s.sharedram >>= bitcount;
                s.bufferram >>= bitcount;
                s.totalswap >>= bitcount;
                s.freeswap >>= bitcount;
                s.totalhigh >>= bitcount;
                s.freehigh >>= bitcount;
        }

        memset(&s_32, 0, sizeof(s_32));
        s_32.uptime = s.uptime;
        s_32.loads[0] = s.loads[0];
        s_32.loads[1] = s.loads[1];
        s_32.loads[2] = s.loads[2];
        s_32.totalram = s.totalram;
        s_32.freeram = s.freeram;
        s_32.sharedram = s.sharedram;
        s_32.bufferram = s.bufferram;
        s_32.totalswap = s.totalswap;
        s_32.freeswap = s.freeswap;
        s_32.procs = s.procs;
        s_32.totalhigh = s.totalhigh;
        s_32.freehigh = s.freehigh;
        s_32.mem_unit = s.mem_unit;
        if (copy_to_user(info, &s_32, sizeof(s_32)))
                return -EFAULT;
        return 0;
}
#endif /* CONFIG_COMPAT */
SYSCALL_DEFINE0(hello_world)
{
    printk(KERN_INFO "Hello World! This is roll number k240874 syscall\n");
    return 0;
}
```

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```
  CC      arch/x86/boot/compressed/acpi.o
  CC      arch/x86/boot/compressed/efi.o
  LD      arch/x86/boot/compressed/vmlinux
  ZOFFSET arch/x86/boot/zoffset.h
  AS      arch/x86/boot/header.o
  CC      arch/x86/boot/main.o
  CC      arch/x86/boot/memory.o
  CC      arch/x86/boot/pm.o
  AS      arch/x86/boot/pmjump.o
  CC      arch/x86/boot/printf.o
  CC      arch/x86/boot/regs.o
  CC      arch/x86/boot/string.o
  CC      arch/x86/boot/tty.o
  CC      arch/x86/boot/video.o
  CC      arch/x86/boot/video-mode.o
  CC      arch/x86/boot/version.o
  CC      arch/x86/boot/video-vga.o
  CC      arch/x86/boot/video-vesa.o
  CC      arch/x86/boot/video-bios.o
  LD      arch/x86/boot/setup.elf
  OBJCOPY arch/x86/boot/setup.bin
  OBJCOPY arch/x86/boot/vmlinux.bin
  BUILD   arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready  (#3)
  CC [M]  fs/efivarfs/efivarfs.mod.o
  CC [M]  .module-common.o
  LD [M]  fs/efivarfs/efivarfs.ko
  CC [M]  drivers/thermal/intel/x86_pkg_temp_thermal.mod.o
  LD [M]  drivers/thermal/intel/x86_pkg_temp_thermal.ko
  CC [M]  net/netfilter/nf_log_syslog.mod.o
  LD [M]  net/netfilter/nf_log_syslog.ko
  CC [M]  net/netfilter/xt_mark.mod.o
  LD [M]  net/netfilter/xt_mark.ko
  CC [M]  net/netfilter/xt_LOG.mod.o
  LD [M]  net/netfilter/xt_LOG.ko
  CC [M]  net/netfilter/xt_MASQUERADE.mod.o
  LD [M]  net/netfilter/xt_MASQUERADE.ko
  CC [M]  net/netfilter/xt_addrtype.mod.o
  LD [M]  net/netfilter/xt_addrtype.ko
  CC [M]  net/ipv4/netfilter/nf_reject_ipv4.mod.o
  LD [M]  net/ipv4/netfilter/nf_reject_ipv4.ko
  CC [M]  net/ipv6/netfilter/nf_reject_ipv6.mod.o
  LD [M]  net/ipv6/netfilter/nf_reject_ipv6.ko
user@DESKTOP-CREJ66G:~/linux-6.19.5$
```

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user@DESKTOP-CREJ66G: ~/linux-6.19.5

```
  CC      arch/x86/boot/memory.o
  CC      arch/x86/boot/pm.o
  AS      arch/x86/boot/pmjump.o
  CC      arch/x86/boot/printf.o
  CC      arch/x86/boot/regs.o
  CC      arch/x86/boot/string.o
  CC      arch/x86/boot/tty.o
  CC      arch/x86/boot/video.o
  CC      arch/x86/boot/video-mode.o
  CC      arch/x86/boot/version.o
  CC      arch/x86/boot/video-vga.o
  CC      arch/x86/boot/video-vesa.o
  CC      arch/x86/boot/video-bios.o
  LD      arch/x86/boot/setup.elf
  OBJCOPY arch/x86/boot/setup.bin
  OBJCOPY arch/x86/boot/vmlinux.bin
  BUILD   arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready  (#3)
  CC [M]  fs/efivarfs/efivarfs.mod.o
  CC [M]  .module-common.o
  LD [M]  fs/efivarfs/efivarfs.ko
  CC [M]  drivers/thermal/intel/x86_pkg_temp_thermal.mod.o
  LD [M]  drivers/thermal/intel/x86_pkg_temp_thermal.ko
  CC [M]  net/netfilter/nf_log_syslog.mod.o
  LD [M]  net/netfilter/nf_log_syslog.ko
  CC [M]  net/netfilter/xt_mark.mod.o
  LD [M]  net/netfilter/xt_mark.ko
  CC [M]  net/netfilter/xt_LOG.mod.o
  LD [M]  net/netfilter/xt_LOG.ko
  CC [M]  net/netfilter/xt_MASQUERADE.mod.o
  LD [M]  net/netfilter/xt_MASQUERADE.ko
  CC [M]  net/netfilter/xt_addrtype.mod.o
  LD [M]  net/netfilter/xt_addrtype.ko
  CC [M]  net/ipv4/netfilter/nf_reject_ipv4.mod.o
  LD [M]  net/ipv4/netfilter/nf_reject_ipv4.ko
  CC [M]  net/ipv6/netfilter/nf_reject_ipv6.mod.o
  LD [M]  net/ipv6/netfilter/nf_reject_ipv6.ko
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano kernel/sys.c
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano arch/x86/entry/syscalls/syscall_64.tbl
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano include/linux/syscalls.h
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano kernel/Makefile
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano include/linux/syscalls.h
user@DESKTOP-CREJ66G:~/linux-6.19.5$ nano Makefile
user@DESKTOP-CREJ66G:~/linux-6.19.5$
```

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